Ruthwik Reddy Sunketa

623-275-8696 | rsunketa@asu.edu | linkedin.com/in/rsunketa

Summary

Computer Engineering PhD student specializing in FPGA architecture, design, and automated EDA flows. Current research focuses on next-generation FPGA architectures for AI/ML workload acceleration and hardware-software co-design. Experience in digital design, verification, physical design, and EDA flow development.

Education

Arizona State University — Tempe, AZ
Ph.D. in Computer Engineering — Aug 2025 – Present
Master's in Computer Engineering — Aug 2023 – July 2025

Relevant Coursework: Reconfigurable Computing, VLSI Design, Hardware Security and Trust, Advanced Hardware and Systems for Machine Learning, Statistical Machine Learning, Real-Time Digital Signal Processing

Research Projects

ATLAS: Automated HLS for DL-Optimized FPGAs — March 2026 – Present

- Developed ATLAS, an automated ML-to-FPGA compilation flow that maps DL workloads onto FPGAs with custom hardblocks without manual RTL design or explicit hardblock instantiation
- Extended hls4ml to lower supported layers into architecture-agnostic GEMM calls; built a GEMM IP generator that composes FPGA hardblocks into arbitrary-shape GEMM IPs and integrates them into HLS as RTL blackboxes
- Evaluated ATLAS across 11 designs on a Tensor Slice-based FPGA, attaining 89% of hand-written RTL compute-area efficiency, a 24% improvement over soft-logic hls4ml, while reducing implementation effort from weeks to hours

3.3/5V Configurable Voltage FPGA — Aug 2025 – Jan 2026

- Explored mixed-voltage FPGA architectures with on-chip Digital LDOs (DLDOs) for drop-in replacement of legacy 3.3/5 V ICs
- Designed and evaluated a custom FPGA fabric using OpenFPGA, including architecture selection, scalable fabric generation, and bitstream-based functional verification
- Performed RTL-to-GDS physical design using OpenLane, including power analysis, DLDO sizing, DRC/LVS sign-off, and tapeout submission on SKY130 technology

OpenFPGA-NoC: Automated Fabric and Bitstream Generation for NoC-based FPGAs — Aug 2024 - June 2025

- Extended OpenFPGA framework to support hardened Network-on-Chips (NoCs) with arbitrary physical topologies
- Developed a dedicated NoC bitstream generator and configuration mechanism to manage dynamic compile-time address translation and global routing mode selection
- Evaluated NoC-based architectures through an automated RTL-to-bitstream flow, confirming a 3.66× frequency gain and 15% lower routing congestion for I/O-intensive workloads over traditional fabrics

Publications

[1] **Ruthwik Sunketa** and Aman Arora, ***ATLAS: Automated HLS for DL-Optimized FPGAs***, FPT, 2026.[Poster, Accepted]

[2] Jiajun Hu, **Ruthwik Reddy Sunketa**, Lei Zhao, Archit Gajjar, Luca Buonanno, and Aman Arora, ***NIFA: Nonlinear IMC Enhanced FPGA for Efficient ML Inference***, FPL, 2026. [Conference]

[3] **Ruthwik Reddy Sunketa**, Jeevesh Choudhury, Aman Arora, ***Programming Domain-Specific FPGA Hardblocks from HLS: An RTL Blackbox Approach***, RAW-IPDPS, 2026. [Workshop]

[4] Archit Gajjar, **Ruthwik Reddy Sunketa**, Lei Zhao, Omar Eldash, Aishwarya Natarajan, Giacomo Pedretti, Aman Arora, Paolo Faraboschi, Jim Ignowski, and Luca Buonanno, ***Azure-Lily: An FPGA Architecture with Analog IMC Engines for Efficient AI***, ACM TACO, 2026. [Journal]

[5] **Ruthwik Reddy Sunketa** and Aman Arora, ***Closing the Loop on FPGA Verification: An Iterative Framework for Maximizing Routing Resource Coverage***, ACM ISFPGA, 2026. [Poster]

[6] **Ruthwik Reddy Sunketa**, Muhammad Ali Farooq, Ganesh Gore, Allen Boston, Pierre-Emmanuel Gaillardon, and Aman Arora, ***OpenFPGA-NoC: Automated Fabric and Bitstream Generation for NoC-based FPGAs***, ACM TRETS, 2025. [Journal]

Technical Skills

Programming: Python, C, C++, Verilog, SystemVerilog, Embedded C

Technologies: Quartus & Xilinx FPGA tools, Synopsys tools (VCS, Design Compiler, PrimeTime), Cadence tools (Genus, Innovus, Virtuoso), Open-source EDA tools (VTR/VPR, OpenFPGA, OpenROAD/OpenLane, Yosys)

Additional Experience

Teaching Assistant: CEN 598 - **Advanced Digital Design and Verification**, ASU — Summer 2025 and Fall 2026
Teaching Assistant: CSE 325 - **Embedded Microprocessor Systems**, ASU — Fall 2024
Intern: Systems Validation - **Medtronic**, Hyderabad, India — Spring 2023